

Braided Operads

Franziska Ahrens

05.08.2025

Outline

- 1 Introduction
- 2 Operads
- 3 Braided Operads

1 Introduction

2 Operads

3 Braided Operads

Motivation

Theorem 1.1 (Recognition principles for E_n -operads [Fie99])

Let A be an operad in the category of compactly generated Hausdorff spaces. A is an E_2 -operad if and only if $A(k)$ is connected and the collection of universal covering spaces $\{\tilde{A}(k)\}_{k \in \mathbb{N}}$ forms a B_∞ -operad, i.e.:

- ① *For all $k \in \mathbb{N}$, the space $\tilde{A}(k)$ is contractible.*
- ② *The braid group B_k acts freely on $\tilde{A}(k)$ for each $k \in \mathbb{N}$.*

Goal:

Develop an analogous theory on braided operads as Lurie has given in Higher Algebra (c.f. [Lur17, Section 2]) for symmetric operads.

1 Introduction

2 Operads

3 Braided Operads

Operads

Definition 2.1

An *operad* $\mathcal{O}$ in $\mathcal{C}$ consists of

- a family of objects $(\mathcal{O}(j))_{j \in \mathbb{N}}$
- a unit morphism $\eta: k \rightarrow \mathcal{O}(1)$
- a composition morphism

$$\gamma: \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \rightarrow \mathcal{O}(j),$$

where $\sum_{1 \leq i \leq k} j_i = j$

- an action of S_n on $\mathcal{O}(n)$

such that suitable associativity, equivariance, unitality constraints hold.

Little Disk Operad in **Top**

Let D^2 be the closed unit disk in $\mathbb{R}^2$. A *little disk* is an embedding $f: D^2 \rightarrow D^2$ of the following form:

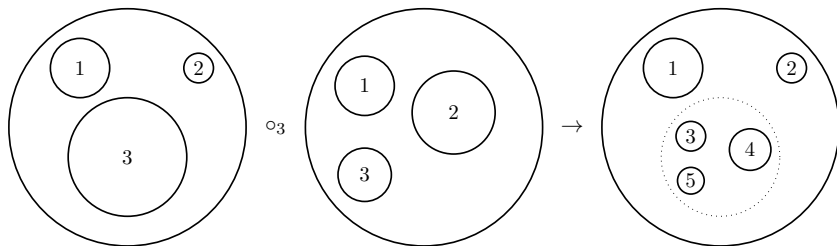
$$f(r_1, \dots, r_n) := (c_1, \dots, c_n) + \lambda(r_1, \dots, r_n)$$

for $(c_1, \dots, c_n) \in D^n$ and $\lambda \in \mathbb{R}$ such that

$$0 < \lambda \leq 1 - \sum_{i=1}^n c_i^2.$$

$$E_2(k) := \{f_1 \sqcup \dots \sqcup f_k : \text{int}(f_i) \cap \text{int}(f_j) = \emptyset\}.$$

Little Disk Operad in **Top**



Definition 2.2

Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of arbitrary categories and $f: x \rightarrow y$ a morphism in $\mathcal{C}$. We call f *cocartesian* if for any morphism $g: y \rightarrow z$ in $\mathcal{C}$ and $\omega: P(x) \rightarrow P(z)$ in $\mathcal{D}$ with $\omega \circ P(f) = P(g)$, there is a unique morphism $\omega': x \rightarrow z$ such that $P(\omega') = \omega$ and $\omega' \circ f = g$. In a diagram this results in the following:

$$\begin{array}{ccc}
 y & \xrightarrow{g} & z \\
 & \searrow f! & \uparrow \exists! \omega' \\
 & & x
 \end{array}
 \quad \xrightarrow{P} \quad
 \begin{array}{ccc}
 P(y) & \xrightarrow{P(g)} & P(z) \\
 & \searrow P(f) & \uparrow \omega \\
 & & P(x)
 \end{array}$$

Definition 2.3

A *colored operad* $\mathcal{O}$ consists of:

- (i) A collection $\{X, Y, Z, \dots\}$ which we call *colors*.
- (ii) For finite collection of colors $\{X_i\}_{i \in I}$, color Y , a set: $\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Y)$ called the set of morphisms.
- (iii) For a finite map $I \rightarrow J$ with fibers $I_j, j \in J$, collections of colors $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$, color Z , a composition map:

$$\prod_{j \in J} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Z).$$

- (iv) A collection of morphisms

$$\{\text{Id}_X \in \text{Mul}_{\mathcal{O}}(\{X\}, X)\}_{X \in \mathcal{O}}$$

called the identity morphisms.

The data needs to fulfill suitable associativity and unitality conditions.

Example 2.4

Let $\mathcal{C}$ be a symmetric monoidal category. By setting for a finite family of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{C}$:

$$\text{Mul}_{\mathcal{C}}(\{X_i\}_{i \in I}, Y) := \text{Hom}(\otimes_{i \in I} X_i, Y),$$

we can define a colored operad $\mathcal{C}$. The composition is given by

$$\prod_{j \in J} \text{Hom}(\otimes_{i \in I_j} X_i, Y_j) \times \text{Hom}(\otimes_{j \in J} Y_j, Z) \rightarrow \text{Hom}(\otimes_{i \in I} X_i, Z)$$

$$((f_j)_{j \in J}, g) \mapsto g \circ (\otimes_{j \in J} f_j)$$

with $\{\text{Id}_X\}_{X \in \mathcal{C}}$ being the unit.

Definition 2.5

The *category of finite pointed sets* $\mathcal{F}in_*$ is defined as follows:

- For each $n \in \mathbb{N}$, we write $\langle n \rangle^\circ := \{1, \dots, n\}$. An objects of $\mathcal{F}in_*$ is $\langle n \rangle := \langle n \rangle_*^\circ$ for all $n \in \mathbb{N}$.
- A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ is a map of finite sets where $\alpha(*) = *$.

Definition 2.6

For $\mathcal{O}$ being a colored operad, the category $\mathcal{O}^{\otimes}$ is defined in the following way:

- i) The objects of $\mathcal{O}^{\otimes}$ are finite sequences of colors of $\mathcal{O}$.
- ii) For $\{X_i\}_{1 \leq i \leq m}$, $\{Y_j\}_{1 \leq j \leq n}$ objects in $\mathcal{O}^{\otimes}$. A morphism $\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$ is given by
 - $\alpha: \langle m \rangle \rightarrow \langle n \rangle$ map in $\mathcal{F}in_*$.
 - family of morphisms

$$\{\phi_j \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \leq j \leq n}$$

in $\mathcal{O}$.

- iii) Composition of morphisms ϕ, ϕ' is determined by the composition in $\mathcal{F}in_*$ and $\mathcal{O}$, i.e. $\bar{\alpha} = \alpha' \circ \alpha$ and $(\phi' \circ \phi)_l \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)$ by applying γ to $((\phi_i)_{i \in \alpha^{-1}(l)}, \phi'_l)$

Let $\mathcal{O}$ be a colored operad. There is a forgetful functor

$$\pi: \mathcal{O}^{\otimes} \rightarrow \mathcal{F}\text{in}_*$$

Proposition 2.7

If $\mathcal{O}$ induces by a symmetric monoidal category $\mathcal{C}$, the following conditions hold for π :

- ❶ **Op-fibration condition:** π is a Grothendieck op-fibration.
- ❷ **Segal condition:** $\mathcal{C} \cong \mathcal{O}_{\langle 1 \rangle}^{\otimes}$ and $\mathcal{O}_{\langle n \rangle}^{\otimes} \cong \mathcal{C}^n$.

Proposition 2.8

$$\{\text{symmetric monoidal categories}\} \cong \left\{ \begin{array}{l} \text{op-fibrations } \mathcal{D} \rightarrow \mathcal{F}\text{in}_* \\ \text{satisfying Segal condition} \end{array} \right\}$$

Symmetric Monoidal Structure on $\mathcal{C} := \mathcal{D}_{\langle 1 \rangle}$

Let $\pi: \mathcal{D} \rightarrow \mathcal{F}\text{in}_*$ a functor satisfying (M1) and (M2).

$\mathcal{D}$	$\mathcal{F}\text{in}_*$
tensor product: $\otimes: \mathcal{C}^2 \cong \mathcal{D}_{\langle 2 \rangle} \rightarrow \mathcal{C}$	$\alpha: \langle 2 \rangle \rightarrow \langle 1 \rangle$ active
braiding: $b_{A,B} A \otimes B \cong B \otimes A$	$\beta: \langle 2 \rangle \rightarrow \langle 2 \rangle$ $1 \mapsto 2$ $2 \mapsto 1.$
associator: $a_{A,B,C}: (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$	$\tau_i^3: \langle 3 \rangle \rightarrow \langle 2 \rangle$ $j \mapsto \begin{cases} j & 1 \leq j \leq i \\ j-1 & i < j \leq n \\ * & j = *. \end{cases}$
tensor unit: $\mathcal{D}_{\langle 0 \rangle} \rightarrow \mathcal{C}$	$\eta: \langle 0 \rangle \rightarrow \langle 1 \rangle$

Definition 2.9

An ∞ -operad is a functor $p: \mathcal{O}^{\otimes} \rightarrow \mathcal{N}(\mathbf{Fin}_*)$ between ∞ -categories satisfying the following conditions:

- (i) For every inert morphism $f: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{N}(\mathbf{Fin}_*)$ and object $C \in \mathcal{O}_{\langle m \rangle}^{\otimes}$, there exists a p -cocartesian morphism $f: C \rightarrow C'$ in $\mathcal{O}^{\otimes}$ lifting f . In particular f induces a functor $f!: \mathcal{O}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{O}_{\langle m \rangle}^{\otimes}$.
- (ii) Let $C \in \mathcal{O}_{\langle m \rangle}^{\otimes}$ and $C' \in \mathcal{O}_{\langle n \rangle}^{\otimes}$ be objects, $f: \langle m \rangle \rightarrow \langle n \rangle$ be a morphism in $\mathbf{Fin}_*$ and let $\mathrm{Map}_{\mathcal{O}^{\otimes}}^f(C, C')$ be the union of connected components of $\mathrm{Map}_{\mathcal{O}^{\otimes}}(C, C')$ which lie over f . Choose p -cocartesian morphisms $C' \rightarrow C'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\mathrm{Map}_{\mathcal{O}^{\otimes}}^f(C, C') \rightarrow \prod_{1 \leq i \leq n} \mathrm{Map}_{\mathcal{O}^{\otimes}}^{\rho^i \circ f}(C, C'_i)$$

is a homotopy equivalence.

- (iii) For every finite collection of objects $C_1, \dots, C_n \in \mathcal{O}_{\langle 1 \rangle}^{\otimes}$, there exists an object $C \in \mathcal{O}_{\langle n \rangle}^{\otimes}$ and a collection of p -cocartesian morphisms $C \rightarrow C_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

1 Introduction

2 Operads

3 Braided Operads

Braided Operads

Definition 3.1

An *operad* $\mathcal{B}$ in $\mathcal{C}$ consists of

- a family of objects $(\mathcal{B}(j))_{j \in \mathbb{N}}$
- a unit morphism $\eta: k \rightarrow \mathcal{B}(1)$
- a composition morphism

$$\gamma: \mathcal{B}(k) \otimes \mathcal{B}(j_1) \otimes \cdots \otimes \mathcal{B}(j_k) \rightarrow \mathcal{B}(j),$$

where $\sum_{1 \leq i \leq k} j_i = j$

- an action of B_n on $\mathcal{B}(n)$

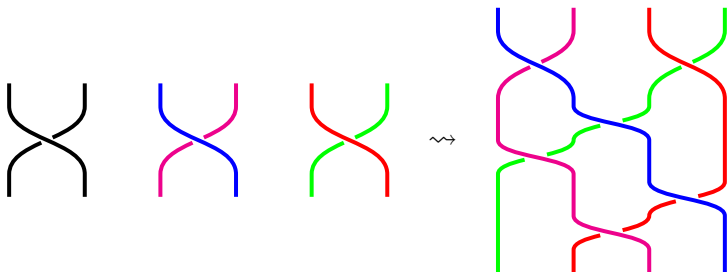
such that suitable associativity, equivariance, unitality constraints hold.

Example: Braid Operad

Let $\mathcal{B}$ be the *braid operad* defined by $\mathcal{B}(n) := B_n$, where B_n denotes the braid group on n strands. The composition morphism

$$\gamma: B_k \times B_{j_1} \times \cdots \times B_{j_k} \rightarrow B_j, \quad (\tau, (\sigma_i)_{1 \leq i \leq k}) \mapsto \tau(\sigma_1 \oplus \cdots \oplus \sigma_k),$$

where $\tau(\sigma_1 \oplus \cdots \oplus \sigma_k)$ denotes the braid that τ -braids k many blocks.



The associativity and unitality conditions follow from the group conditions.

- Another example is the universal cover operad of E_2 . Consider the unordered little disk space E_2/S . By Fox and Neuwirth (c.f. [FN62]): the fundamental group of $E_2(n)/S_n$ is the n -th braid group B_n .
- For any finite set I , we consider the little disks in $E_2(n)$ as labeled by I instead of $\{1, \dots, n\}$, called I -labeled configuration.

Definition 3.2

The **Grpd**-enriched category **BR** of *braided sets*:

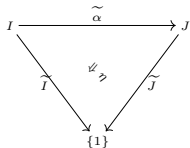
- (i) Objects are finite sets.
- (ii) A morphism $\tilde{\alpha}: I \rightarrow J$ in **BR** consists of
 - a map of finite sets $\alpha: I \rightarrow J$
 - For all $j \in J$ α_j -labeled configuration $\{\widetilde{\alpha_j}\}$.
- (iii) Composition consists of composing maps of finite sets and "inserting disks", analogously to the composition in the little disk operad E_2 .
- (iv) The identity morphism Id_I is given by the identity on the finite set I and the collection of identity configurations $\text{Id}^{(i)}$.
- (v) For morphisms $\tilde{\alpha}, \tilde{\beta}: I \rightarrow J$ in **BR** with the same underlying map $\alpha = \beta$, a 2-morphism in **BR** consists of a family of homotopy classes of paths, $j \in J$:

$$\eta_j: \widetilde{\alpha^{-1}(j)} \Rightarrow \widetilde{\beta^{-1}(j)}.$$

Definition 3.3

A *colored braided operad* $\mathcal{B}$ consists of the following data:

- (i) A collection $\{X, Y, Z, \dots\}$ of colors.
- (ii) For I -labeled configuration $\widetilde{I}$, collection of colors $\{X_i\}_{i \in I}$, color Y , a set $\text{Mul}_{\mathcal{B}}^{\widetilde{I}}(\{X_i\}_{i \in I}, Y)$, called the set of morphisms.
- (iii) For every triangle in BR:



along with collections of colors $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$ and color Z , a composition map:

$$\gamma: \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\alpha_j}(\{X_i\}_{i \in \alpha_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\widetilde{J}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{B}}^{\widetilde{I}}(\{X_i\}_{i \in I}, Z).$$

- (iv) For any I -labeled configuration $\widetilde{I}$, $|I| = 1$, color X , an identity morphism on X : $\text{Id}_X \in \text{Mul}_{\mathcal{B}}^{\widetilde{I}}(\{X\}, X)$

The data satisfies associativity and unitality constraints.

Definition 3.4

The category of braided pointed sets, $\mathbf{BR}_*$, is the following **Grpd**-enriched category:

- i) The objects of $\mathbf{BR}_*$ are finite sets with a distinguished point $*$.
- ii) A morphism $\tilde{\alpha}_*: I \rightarrow J$ consists of a map $\alpha: I \rightarrow J$ with $\alpha(*) = *$ together with a morphism in $\mathbf{BR}$:

$$\tilde{\alpha}: \alpha^{-1}(J \setminus \{*\}) \rightarrow J \setminus \{*\}$$

- iii) For morphisms $\tilde{\alpha}_*, \tilde{\beta}_*: I \rightarrow J$ with the same underlying map $\alpha = \beta$, a 2-morphism $\eta_*: \tilde{\alpha}_* \rightarrow \tilde{\beta}_*$ corresponds to a 2-morphism $\eta: \tilde{\alpha} \Rightarrow \tilde{\beta}$ in $\mathbf{BR}$.

Definition 3.5

Let the category of standard braided pointed sets $\mathbf{Br}_*$ be the full subcategory of $\mathbf{BR}_*$ on the objects $\langle n \rangle$, $n \in \mathbb{N}$.

Definition 3.6

Let $\mathcal{B}$ be a colored braided operad. Define the **Grpd**-enriched category $\mathcal{B}^\otimes$:

- (i) Objects are finite sequences $\{X_i\}_{1 \leq i \leq n}$ of colors of $\mathcal{B}$.
- (ii) For objects $\{X_i\}_{1 \leq i \leq m}$, $\{Y_j\}_{1 \leq j \leq n}$, a morphism

$$\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$$

given by $\tilde{\alpha}: \langle m \rangle \rightarrow \langle n \rangle$ in $\mathcal{Br}_*$ together with for $1 \leq j \leq n$:

$$\phi_j \in \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}^j}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)$$

in $\mathcal{B}$. We call $\tilde{\alpha}$ the underlying morphism of ϕ .

- (iii) Composition of morphisms is determined by the composition in $\mathcal{Br}_*$ and $\mathcal{B}$.
- (iv) For morphisms ϕ, ψ , a 2-morphism $\bar{\eta}: \phi \rightarrow \psi$ consist of a 2-morphism

$$\bar{\eta}: \tilde{\alpha}_\phi \Rightarrow \tilde{\alpha}_\psi$$

in $\mathcal{Br}_*$ such that $\gamma((\text{Id}_{X_i})_{i \in \alpha_{\phi_j}}, \psi_j) = \phi_j$ for γ corresponding to $\bar{\eta}_j$.

Let $\mathcal{B}$ be a colored braided operad. There is a forgetful functor

$$\pi: \mathcal{B}^{\otimes} \rightarrow \mathcal{B}r_*$$

Proposition 3.7

For π satisfies the following properties:

M1 *Op-fibration condition:* π is a Grothendieck op-fibration.

M2 *Segal condition:* $\mathcal{C} \cong \mathcal{B}_{\langle 1 \rangle}^{\otimes}$ and $\mathcal{B}_{\langle n \rangle}^{\otimes} \cong \mathcal{C}^n$.

we can construct a braided monoidal structure on $\mathcal{C}$.

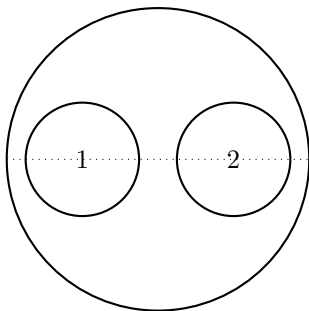
We define $\otimes: \mathcal{C}^2 \rightarrow \mathcal{C}$, by lifting the following morphism in $\mathcal{B}r_*$:

$$\mu: \langle 2 \rangle \rightarrow \langle 1 \rangle$$

$$1 \mapsto 1$$

$2 \mapsto 1$

with configuration $b_2 \in E_2(2)$:

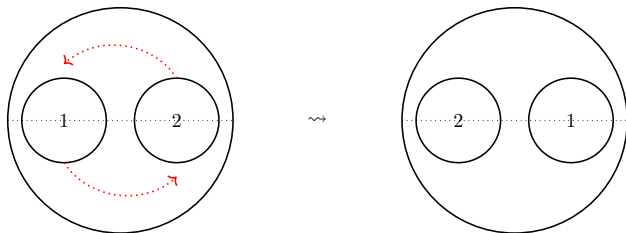


Let $\beta: \langle 2 \rangle \rightarrow \langle 2 \rangle$ with $\beta(1) = 2, \beta(2) = 1$ denote the morphism in $\mathcal{B}r_*$ with the trivial configurations $(\text{Id}^2, \text{Id}^1)$.

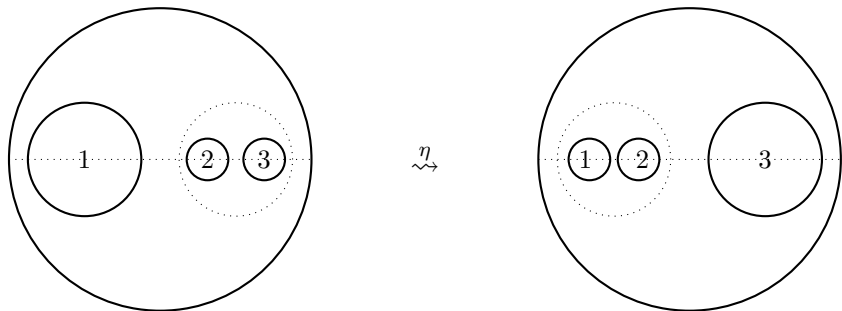
We need to choose a homotopy class of paths for:

$$\begin{array}{ccc} \langle 2 \rangle & \xrightarrow{\beta} & \langle 2 \rangle \\ \downarrow \mu & & \downarrow \mu \\ \langle 1 \rangle & \dashrightarrow & \langle 1 \rangle \end{array}$$

We choose the half twist:



Similarly, for the associator, we choose a path along the horizontal line:



This defines in fact a braided monoidal structure on $\mathcal{C}$.

Operadic Fibration - Definition (1/2)

Definition 3.8 (Operadic Fibration)

An *operadic fibration* is a functor between **Grpd**-enriched categories

$$\pi: \mathcal{D} \rightarrow \mathcal{B}r_*$$

satisfying the following conditions:

- (i) The functor π is a categorical fibration.
- (ii) For every inert morphism $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{B}r_*$ and object $X \in \mathcal{D}$ there exists a π -cocartesian morphism $f: X \rightarrow X'$ lifting $\tilde{\alpha}$.
- (iii) For a morphism $\tilde{\alpha}$ in $\mathcal{B}r_*$, the groupoid of morphisms lying over $\tilde{\alpha}$:

$$\mathrm{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$$

is discrete.

(continued on next slide...)

Operadic Fibration - Definition (2/2)

Definition 3.9 (Operadic Fibration (continued))

- ⓓ Let $X \in \mathcal{D}_{\langle n \rangle}$ and $X' \in \mathcal{D}_{\langle m \rangle}$ be objects, $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\text{Br}_\times$ and let

$$\text{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$$

be the morphisms which lie over $\tilde{\alpha}$. Choose π -cocartesian morphisms $X' \rightarrow X'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\text{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X') \rightarrow \prod_{1 \leq i \leq n} \text{Hom}_{\mathcal{D}}^{\rho^i \circ \tilde{\alpha}}(X, X'_i)$$

is an equivalence of groupoids.

- ⓓ For every finite collection of objects $X_1, \dots, X_n \in \mathcal{D}_{\langle 1 \rangle}$, there is an object $X \in \mathcal{D}_{\langle n \rangle}$ and a collection of π -cocartesian morphism $X \rightarrow X_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

Proposition 3.10

Let $\mathcal{B}$ be a colored braided operad and π the forgetful functor

$$\begin{aligned}\pi: \mathcal{B}^{\otimes} &\rightarrow \mathcal{B}r_* \\ [X_1, \dots, X_n] &\mapsto \langle n \rangle \\ (f: [X_1, \dots, X_n] \rightarrow [X'_1, \dots, X'_m]) &\mapsto (\alpha_f: \langle n \rangle \rightarrow \langle m \rangle) \\ (\eta: \psi \Rightarrow \phi) &\mapsto (\tilde{\eta}: \alpha_{\phi} \Rightarrow \alpha_{\psi})\end{aligned}$$

where α_f refers to the morphism in $\mathcal{B}r_*$ underlying f and $\tilde{\eta}$ the 2-morphism underlying η .

Then, π is an operadic fibration.

Proposition 3.11

Let

$$\pi: \mathcal{D} \rightarrow \mathcal{B}r_*$$

be an operadic fibration over $\mathcal{B}r_*$. Then we can reconstruct a colored braided operad $\mathcal{B}$ from $\mathcal{D}$ such that, $\mathcal{D} \cong \mathcal{B}^\otimes$ and the functor

$$\mathcal{B}^\otimes \cong \mathcal{D} \xrightarrow{\pi} \mathcal{B}r_*$$

is the usual forgetful functor from $\mathcal{B}^\otimes$ to $\mathcal{B}r_*$.

Thank you for your attention!